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Roll No

AU-5004-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Automotive Chassis System

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume suitable missing data if any.

1. a) Draw the layout of conventional chassis with a neat diagram and explain about various parts on it.
b) How is a frame different from a chassis? Discuss the design aspects and salient features of frame.
2. a) Why are stub axles fitted in front axles? Sketch the different types of stub axles.
b) State the functions and types of steering gears. Explain the construction and working of rack and pinion steering gears.
3. a) What do you mean by the term “Independent Suspension System”?
b) What are the functions of a shock absorber? Describe the telescopic shock absorber with a sketch.
4. a) Discuss the classification of brakes for vehicles. What are the advantages and disadvantages of disc brakes compared to drum brake?
b) Differentiate between a leading shoe and a trailing shoe.

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5. a) Differentiate between the conventional tube type tyre and tubeless tyre.
b) Discuss wheel balancing and wheel alignment.
6. a) Explain chassis overhang and ground clearance.
b) What is turning radius, reversible steering and irreversible steering? What is its significant?
7. a) Explain the characteristics, constructional features and advantages of frameless body.
b) What do you understand by understeering and oversteering?
8. a) Compare spoke wheel with disc wheel.
b) Describe briefly the castor and camber angle.

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